Optimizing Automotive Seating Position to Reduce Injury Severity in Short-Statured Drivers

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Project Goals

Part One: Injury Simulations and Optimizing Seating Position

This project's goal is to find the optimal seating position for short-statured drivers that reduces the severity of injury in a frontal collision by running a multi-physics in-silica simulation using LS-DYNA.



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Part Two: Optimal Seating Position Verification System

The previous conclusions regarding seating position will then be leveraged to create a system that will guide a driver into their proper position based on stature. Sensors will collect the relevant distance and rotational measurements and it will be run through an algorithm that will consider the driver's measurements, determine their ideal position, and check if each measurement is within a proper and safe range.

Optimization Procedure

Build a Sled Model

- Automobile "neutral" with a seat, seatbelt, airbag, and steering column
- Driver position
- For initial simulations, 50th FE Rigid Dummy
- Wearing seatbelt
- Frontal collision
- Impact at 40mph
- Airbag deployed

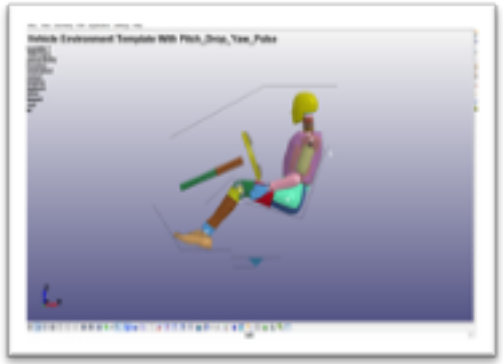
Measure Human Participants

- Collect length and angle measurements to describe seating position according to *Driver Seat: Relative Position Measurements Chart*



Run Sled Simulation

- Use human participant data to set positioning for VATD in the crash and execute sled simulation



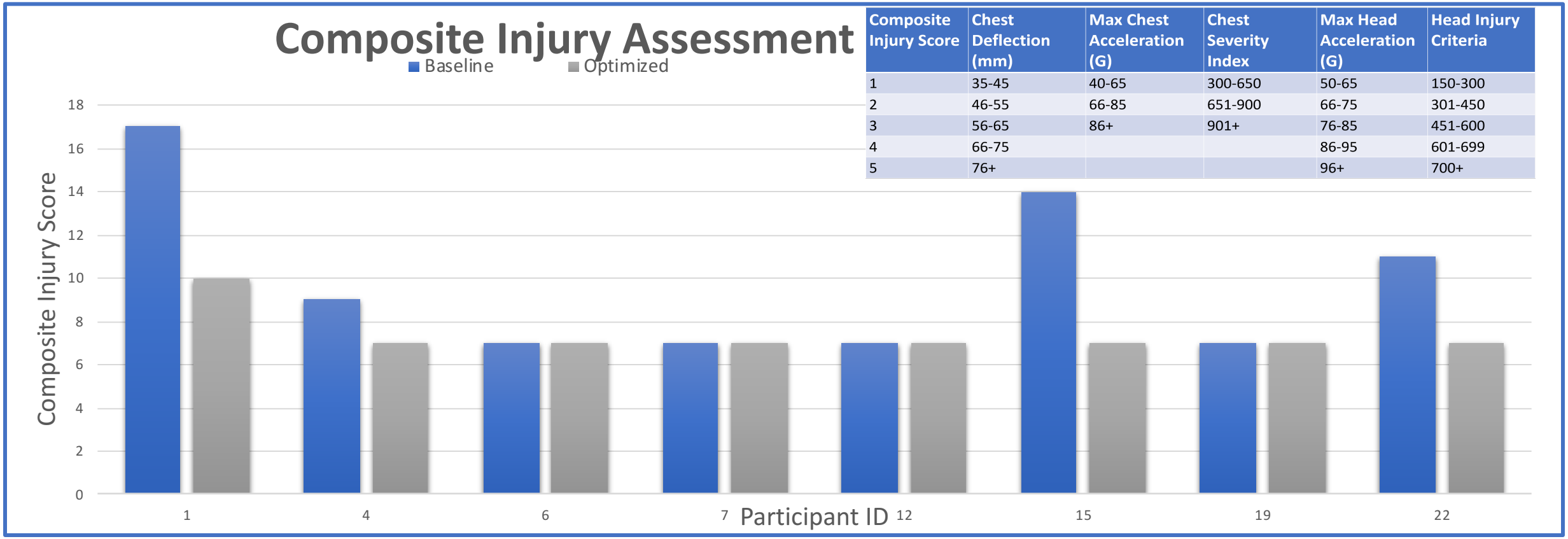
Post-Process Simulation

- Perform analysis to determine chest deflection (mm), maximum chest acceleration (G), Chest Severity Index, head acceleration (G), and Head Injury Criteria from the mathematical outputs of the simulation

Optimize Seating Position

- Identify subset of high-injury, short-stature occupant models
- Adjust parameters to minimize injury severity

Simulation Data: Composite Injury Score



Composite Injury Score	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria
1	35-45	40-65	300-650	50-65	150-300
2	46-55	66-85	651-900	66-75	301-450
3	56-65	86+	901+	76-85	451-600
4	66-75			86-95	601-699
5	76+			96+	700+

The Composite Injury Score (CIS) was developed to easily classify total injury severity by adding up scores from the different parameters using a tiered point system.

Optimization Conclusions

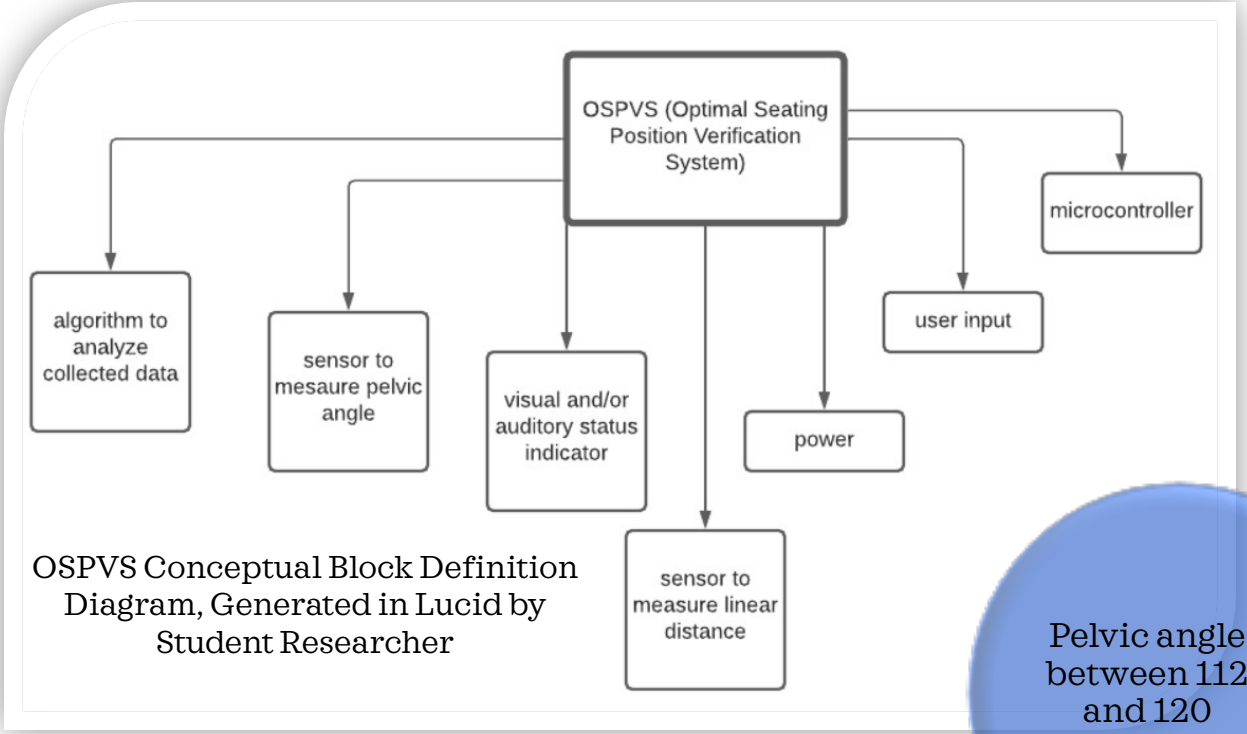


A total of 145 simulations were run, and after performing optimization simulations to all short-statured participants, changes to the pelvic angle, specifically a pelvic angle between 112 and 120 degrees, while keeping the chest distance as close as possible to the baseline, proved to have the most significant positive change on the injury severity scores.

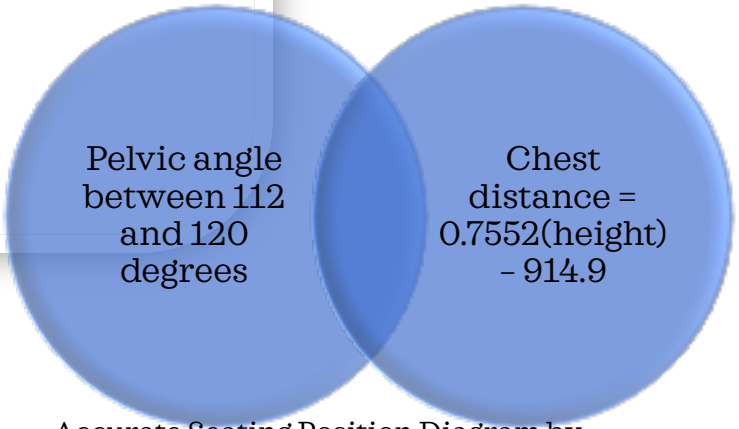
For the drivers whose optimized positions exhibited improved injury severity scores, the overall effect was a 39% reduction in severity.

Part Two:
Optimal Seating Position Verification System

Requirements and Optimization Algorithm Rules

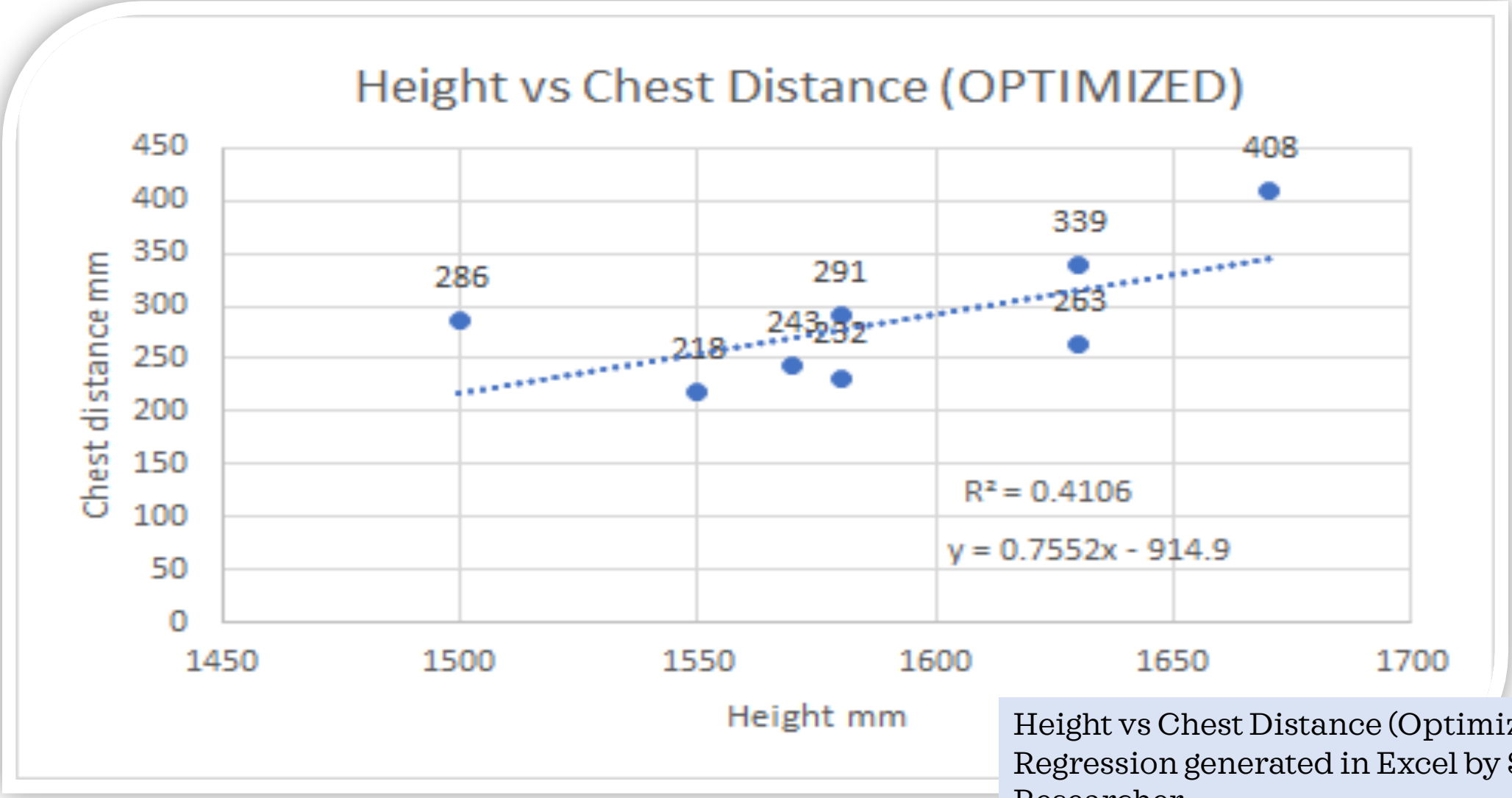


OSPVS Conceptual Block Definition Diagram, Generated in Lucid by Student Researcher



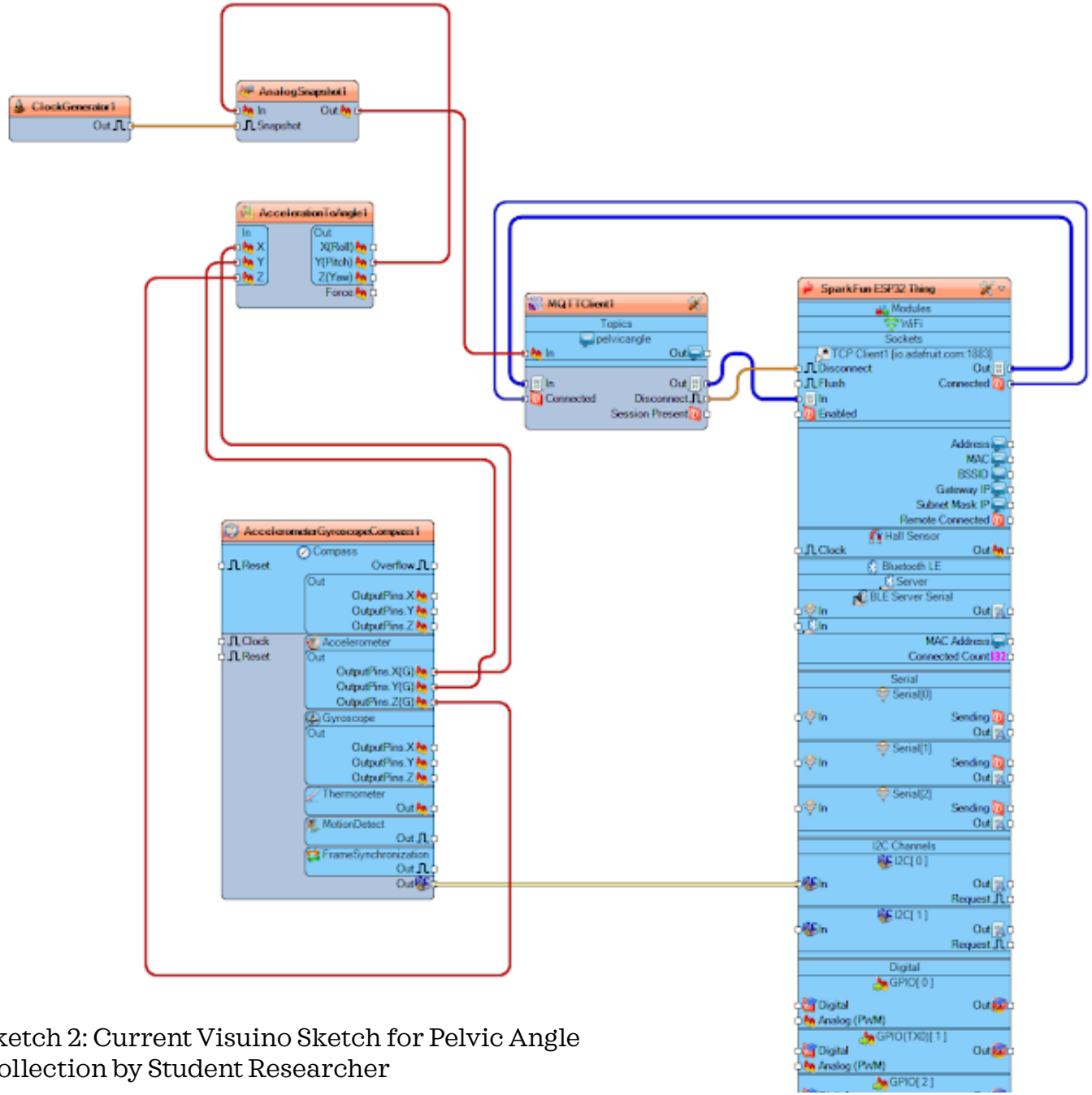
Accurate Seating Position Diagram by Student Researcher

Height vs. Chest Distance Regression



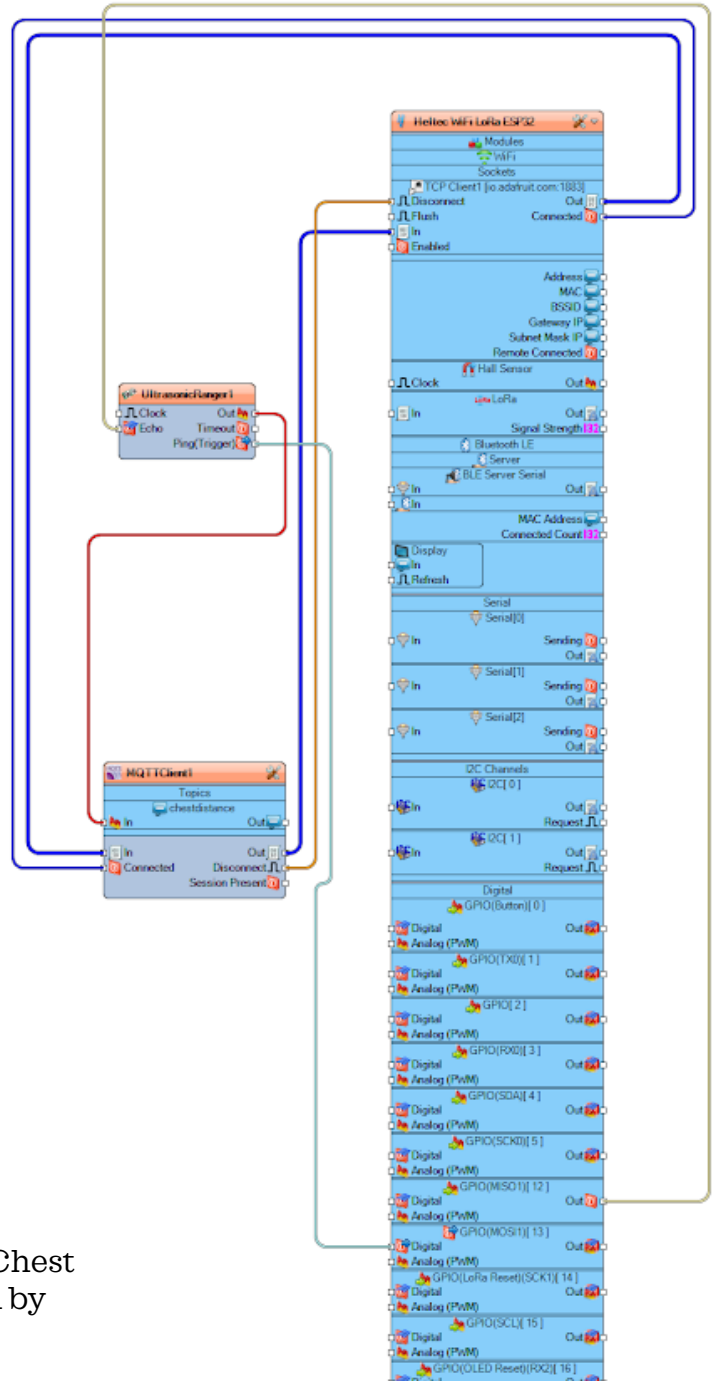
Height vs Chest Distance (Optimized) Regression generated in Excel by Student Researcher

OSPVS Prototype Visuino Sketches: Pelvic Angle Collection



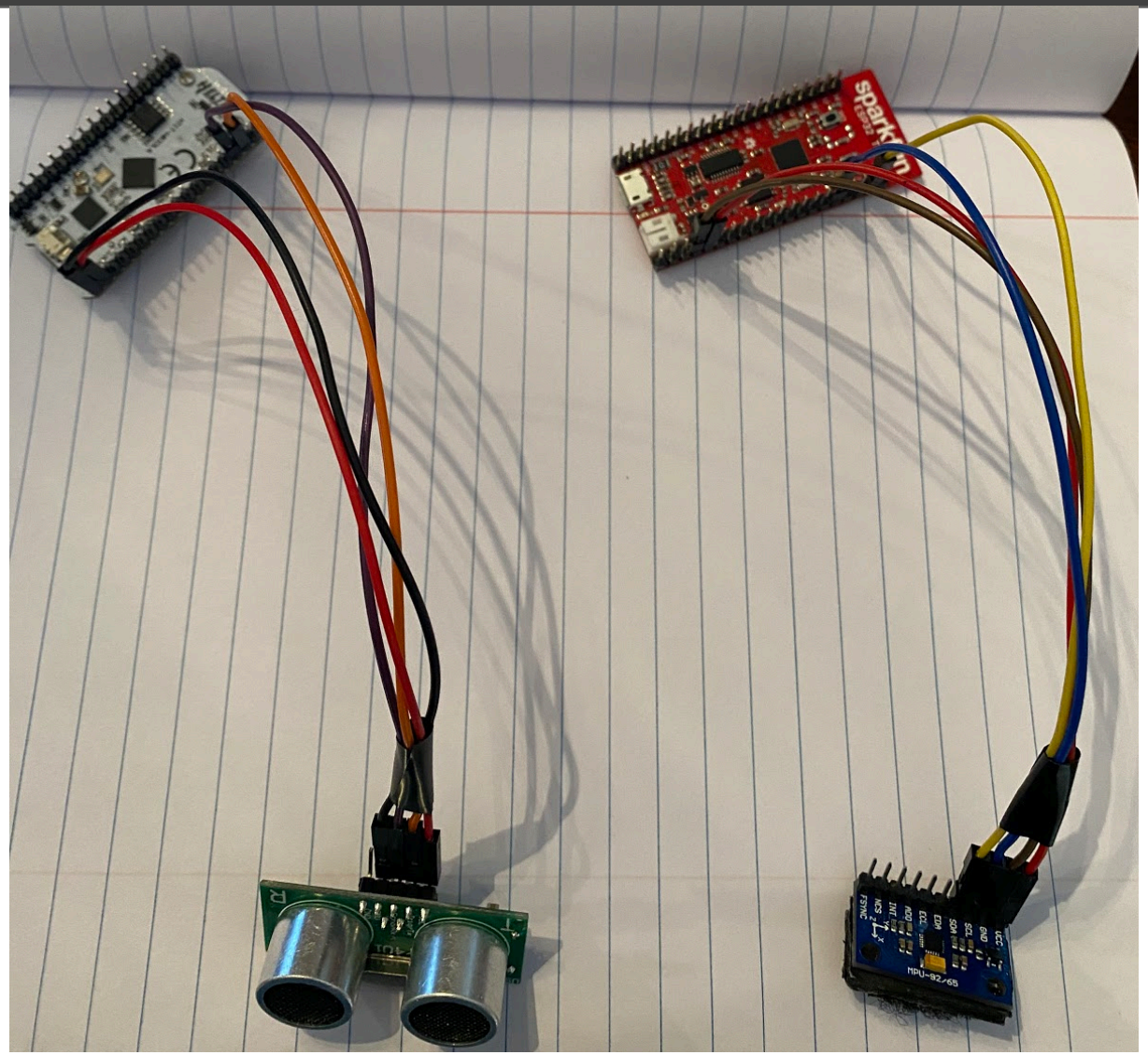
Sketch 2: Current Visuino Sketch for Pelvic Angle Collection by Student Researcher

OSPVS Prototype Visuino Sketches: Chest Distance Collection



Sketch 3: Current Visuino Sketch for Chest Distance Collection by Student Researcher

OSPVS Physical Prototype



Implementation Process

Place box of Chest Device in center of chest on the seatbelt

Place Pelvic Angle Device box on seat itself

Launch the app

Input client height and biological sex

Algorithm will determine ideal chest distance range

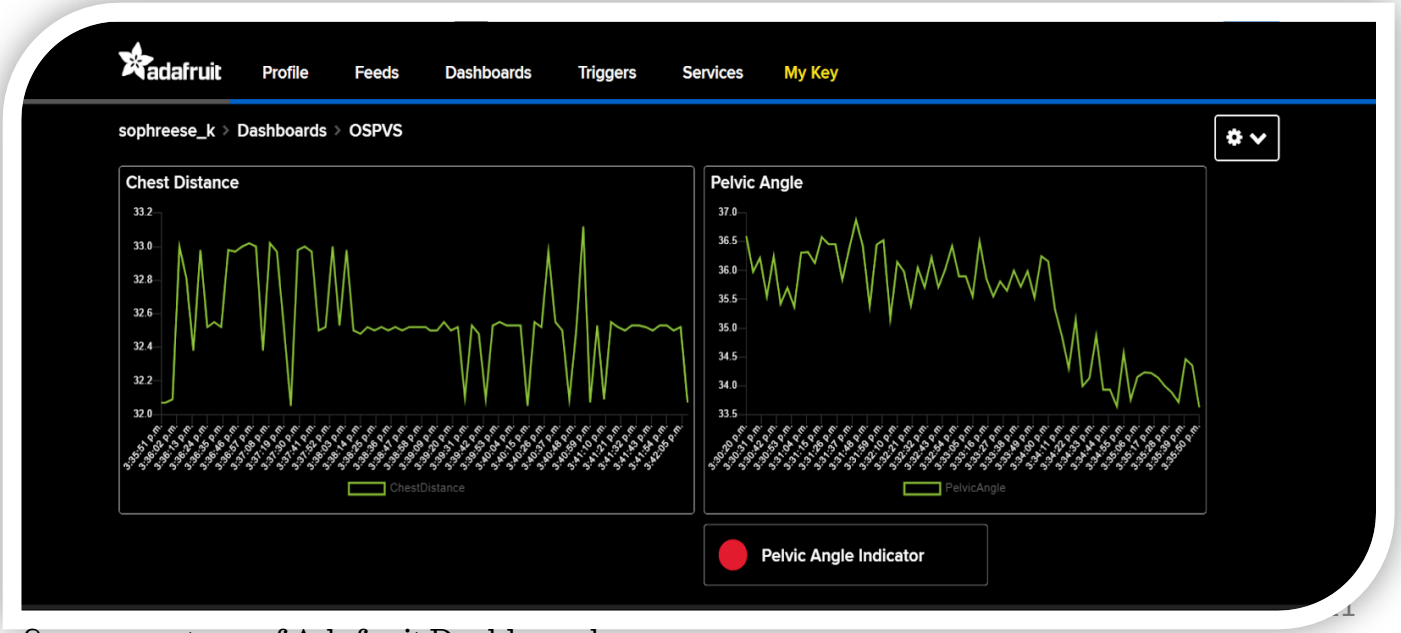
App will prompt driver to begin adjusting their seat

Chest Distance indicator will turn green when it is in proper range

Pelvic Angle indicator will turn green when in proper range

When both indicators are green, the seat is in a proper position

Remove sensors



Screencapture of Adafruit Dashboard

Error Analysis



Human Response/Personal Preference

- No simulation can accurately include how a human would respond to a car crash, as likely they would know it is coming and try to take measures against that.
- Humans are also likely to simply have personal preference when it comes to their seating positions.



Human Error in Measurement Collection

- There is the likely possibility that error was introduced into the data through human error in data collection.
- This, however, would only have marginal effects on the data.



Insufficient Model Fidelity

- Since a student edition was run, the model is limited to 10,000 finite elements, which might have limited the accuracy of the data results.
- However, this would only produce marginal changes and not change the statistical significance.
- Also, only ⅔ of the main measurements were exactly matched.

Acknowledgements

Enrique Krajmalnik

- Thank you to my mentor, Enrique Krajmalnik, for assisting me with finding and understanding literature, learning to use LS-DYNA, working with Visuino, and running simulations. He also worked as a sounding board, helping me work through ideas and answer questions.

Invisible Women: Exposing Data Bias in a World Designed for Men by Caroline Criado-Perez

- Thank you to this book for introducing me to the world of data bias and, in addition to my own experiences, inspiring the development of this research project.

Applications and Future Plans

- Get more human data and complete the LS-DYNA simulations with additional VATD bodies and rigidity
- Design the algorithm and apply it to the Chest Distance indicator in the cloud/application
- Plot the height vs chest distance regression again, accounting for differences in biological sex (two separate regressions)
- Mount prototype in vehicle, test, and validate results
- Get the automotive manufacturers to listen
- Create system to automatically readjust seat based on safety data

The Problem: Injury Statistics

Short-statured, out-of-position drivers have a 47% increase in likelihood of serious injury and a 17% increase in likelihood of death. (Criado-Perez, C. (2019, February 23). The deadly truth about a world built for men – from stab vests to car crashes.)

In the 1950s, 50th percentile male (177 cm tall) crash test dummies were introduced. In 2011, female crash test dummies were introduced, but they were not anthropomorphically correct- they were simply scaled down male models. Even today, no testing requires the use of accurate female dummies. (Criado-Perez, C. (2019, February 23). The deadly truth about a world built for men – from stab vests to car crashes.)

While testing for the average of the population is logical, it endangers the extreme ends of the population. Simply extrapolating or adjusting data is not accurate. (Welsh, R., Morris, A., & Clift, L. (2003). The effect of height on injury outcome for drivers of European passenger cars.)

A Modern Approach: Virtual Anthropomorphic Testing Devices (VATDs)

Problems Posed by Physical Crash Test Dummies

- Expensive
- Single-use
- Materials cannot accurately simulate human responses

Virtual Anthropomorphic Testing Devices (VATDs)

- VATDs are in-silica versions of physical crash test dummies with increased capabilities to be used in virtual crash test simulations
- VATDS offer increased biofidelity over crash test dummies and can simulate a broader group of humans
- This project utilizes pre-existing VATDs whose driving position will be adjusted according to human data

Finite Element Analysis

Method to solve complex math/physics problems

Take a complex structure and decompose into many finite elements

- Finite elements can be solved individually and then reassembled

LS-DYNA

- LS-DYNA is the industry standard for nonlinear finite element analysis simulations for automotive companies in occupant safety and crash test applications
- This software offers extensive material and element libraries and contact algorithms as well as existing open-source models

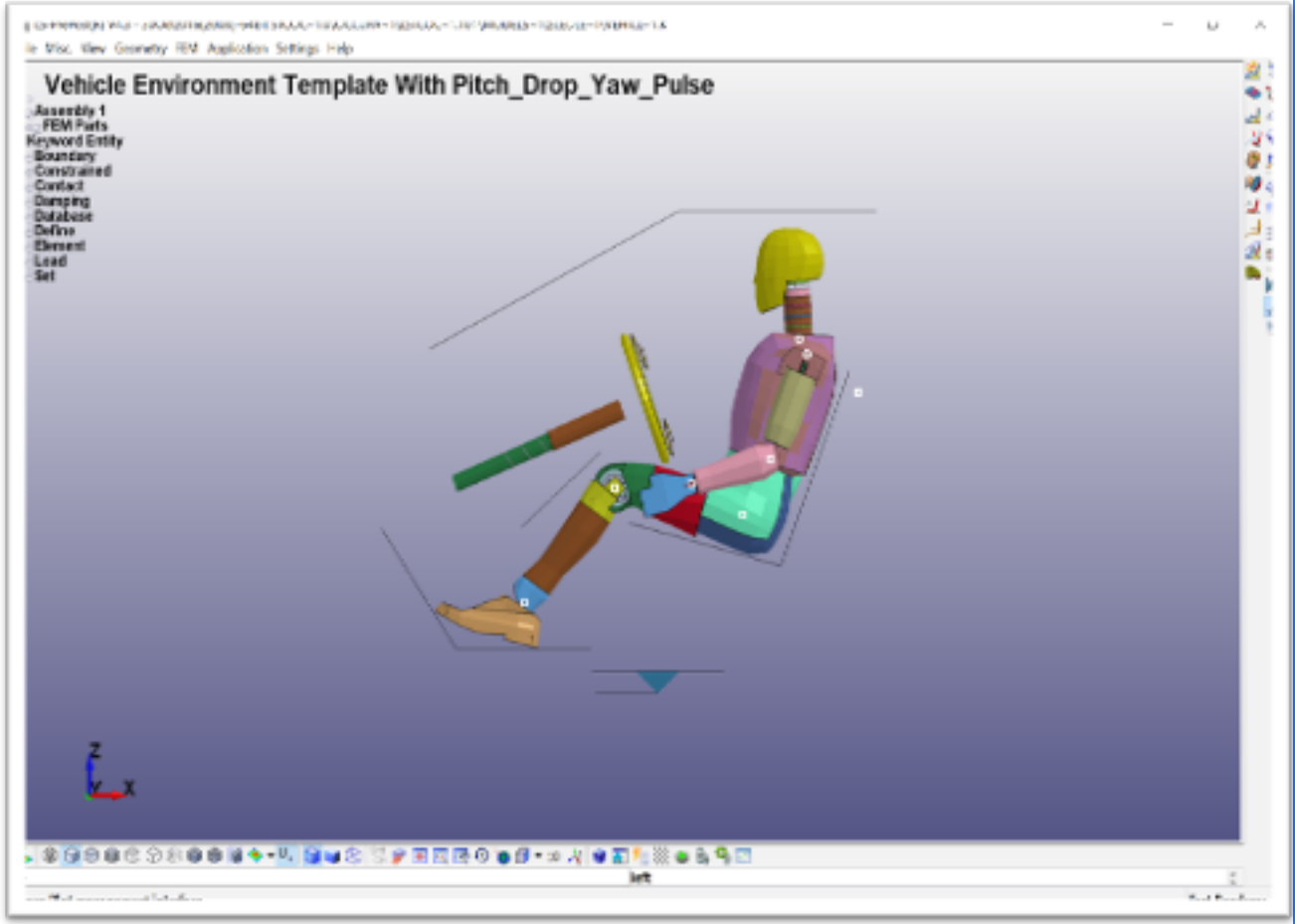
Human Measurements



Driver Seat: Relative Position Measurements Diagram: Image from "Car Side View Person No Airbag No Seat Belt." *123RF;* Measurements and Angles added by Student Researcher; 11/20/2019

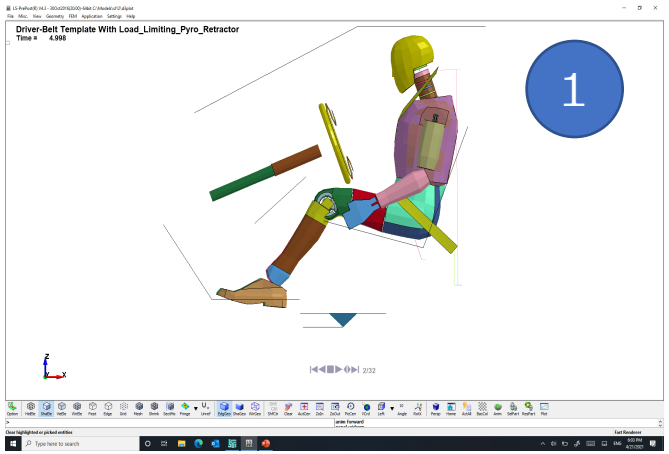
Image of Driver/Data Set 2 (45th Percentile Male) in Driving Position, Taken by Student Researcher, 11/23/2019

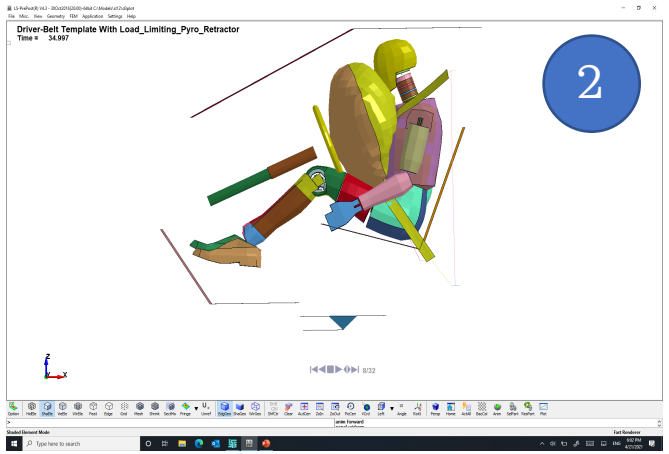


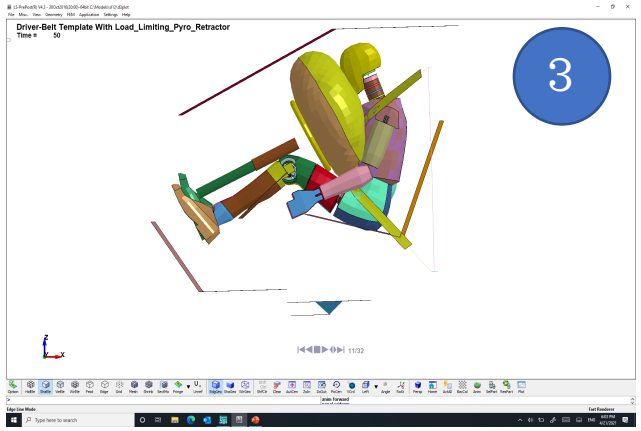


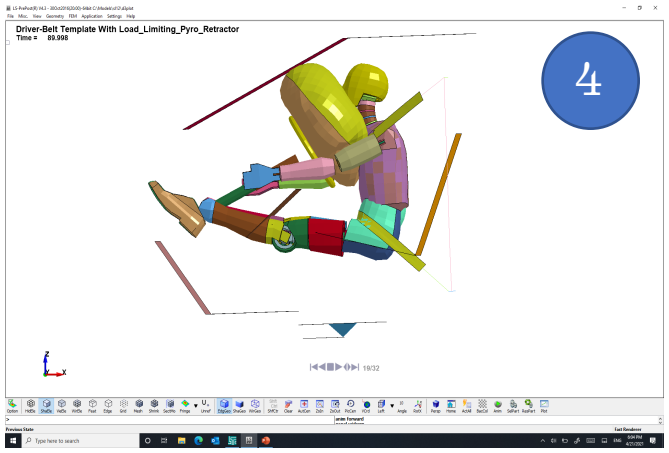
Screen capture of Sled Model with VATD from LS-DYNA; Loaded by Student Researcher, 12/2/2019

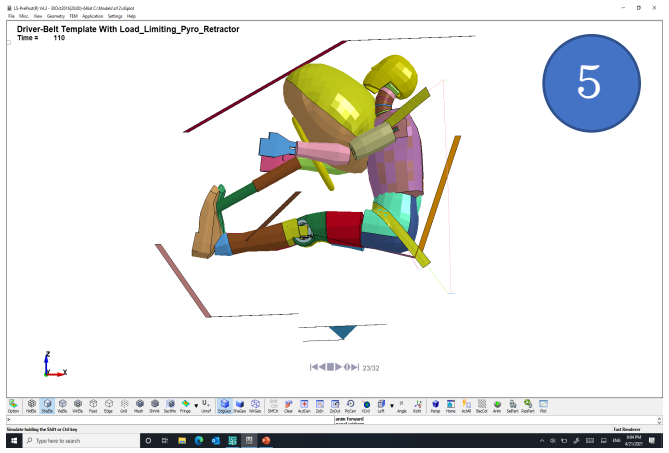
Simulation Sample

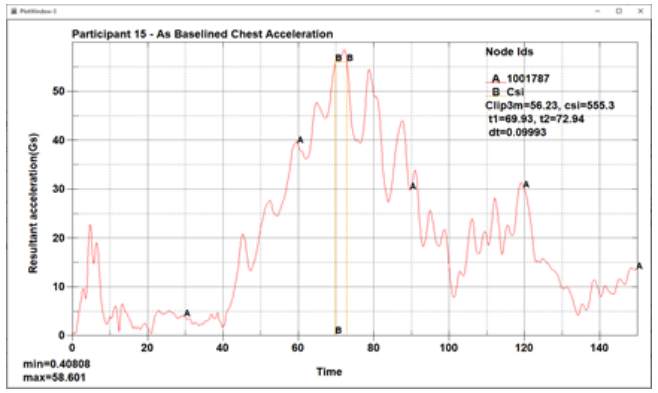












This photo series displays a sample of one crash simulation being run, and the graph in the bottom right corner shows LS-DYNA's graphical output of the simulation, specifically for chest acceleration, which is then analyzed and key information is extrapolated.

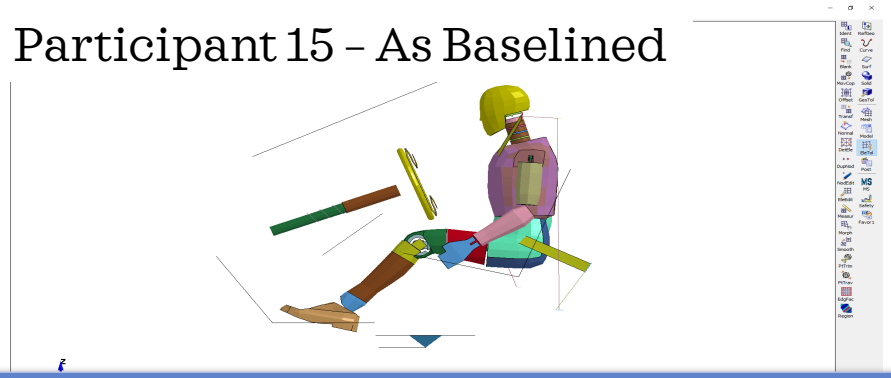
The simulation screenshots and graphical output are from separate simulations of different participants.

Simulation Data

Demographic Data				As Baselined									As Optimized						
Participant ID	Height (cm)	Sex	Age/Height Percentile	B (mm)	C (mm)	Pelvic Angle (deg)	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	**Composite Injury Score**	Method	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
1	167	M	13	445	381	109	38	133	1314	119.9	893	17	B25	37	84	692	65	666	10
4	157	F	26	280	229	108	53	52	487	66.6	561	9	B4D	45	52	451	55	499	7
6	163	F	52	440	280	120	40	44	374	61.3	581	7	F25	39	42	372	60	573	7
7	158	F	26	270	260	118	41	46	360	53.2	560	7	B4D	15	45	505	61	492	7
12	163	F	57	343	292	103	42	52	479	58.7	588	7	PAM	44	38	342	56	542	7
14	170	M	23	470	406	115	51	62	723	84.0	970	13	Sim Error						
15	150	F	24	432	286	94	83	59	555	65.8	700	14	PAM	40	42	363	59	557	7
19	158	F	28	279	248	94	42	59	547	55.9	571	7	B4D	40	44	378	59	568	7
22	155	F	17	343	279	88	49	70	838	60.0	621	11	PAM	45	43	398	63	575	7

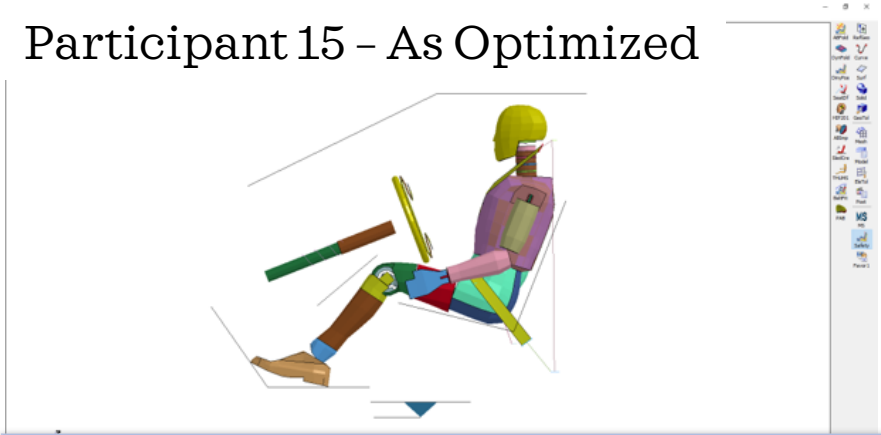
Seating Position Optimization

Participant 15 – As Baselined



Participant ID	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
15	83	59	555	66	700	14

Participant 15 – As Optimized



Participant ID	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
15	40	42	363	59	557	7

Method Name	Description
B25	X translated -25 mm
F25	X translated +25 mm
B4D	Pelvic angle rotated -4 degrees
PAM	Match pelvic angle of seat in initial sled (112 degrees) but adjust chest and head distances to match participant baseline data

Optimization Method Description Chart

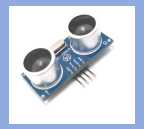
Measurement	As seated	As baselined	As optimized
B mm	432	363	432
C mm	286	286	286
Pelvic angle deg	94	94	112

Part Two:
Optimal Seating Position Verification System

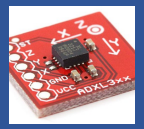
Electronics Background



Microcontrollers are compact integrated circuits designed to perform specific tasks



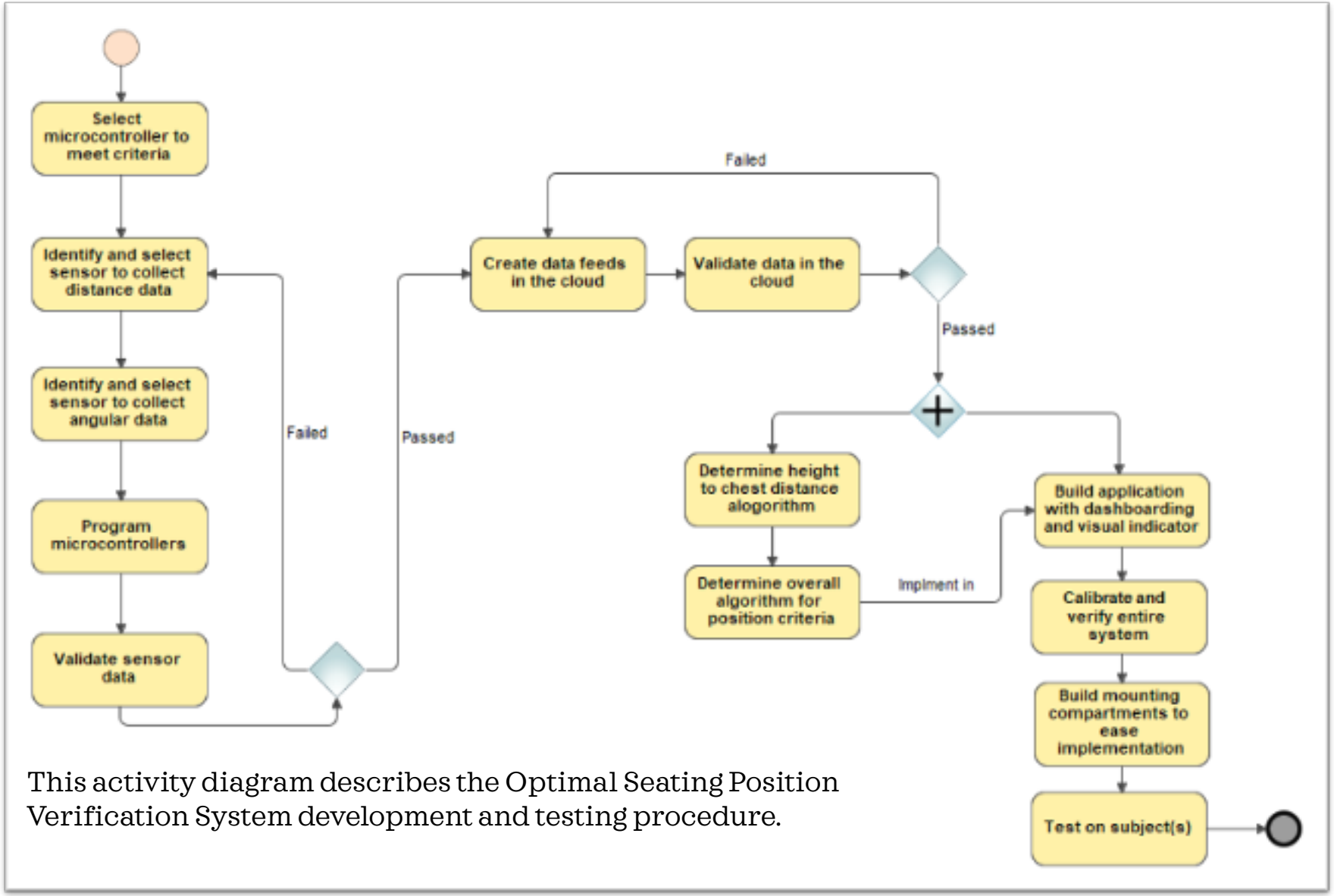
Ultrasonic range finders have two main components: ping, which sends out the sound waves, and echo, which receives the sound waves



Accelerometers collect angular positioning data

MQTT (Message Queue Telemetry Transport) is an IoT (Internet of Things) messaging protocol

OSPVS Development Procedure



This activity diagram describes the Optimal Seating Position Verification System development and testing procedure.

OSPVS Prototype Diagram

